The Photoelectric Effect

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was awarded the Nobel Prize in Physics in 1921.

The photoelectric effect is a phenomenon where light shining on a metal surface ejects electrons from that surface. This effect was one of the key pieces of evidence for the particle-like nature of light.

Historical Background:

In the late 19th and early 20th centuries, scientists observed that when light hits a metal, electrons are emitted. Classical physics, based on wave theory of light, predicted that the energy of ejected electrons should depend on the intensity of light and should be emitted after a certain delay once enough energy accumulated. However, experiments showed something different: electrons were emitted almost instantly when illuminated with light above a certain frequency, regardless of the light's intensity, and the kinetic energy of the emitted electrons depended on the light's frequency, not its intensity.

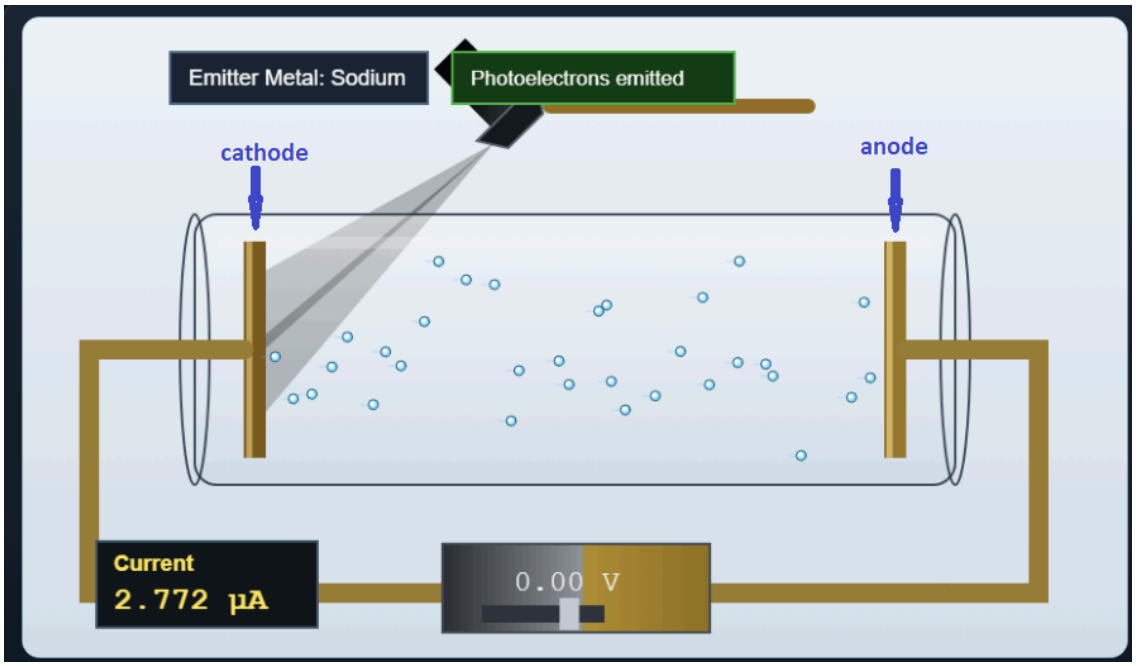
Classical Interpretation and Mistakes:

Classical wave theory suggested that the light's energy was spread evenly over the wavefront, so increasing the intensity should increase the energy transferred to electrons over time, eventually ejecting them. However, this didn't match experimental results. It was also assumed that electrons would need to absorb multiple photons, or that energy would build up gradually until an electron broke free. Both ideas failed to explain the immediate emission of electrons and the dependence of electron emission on frequency.

Einstein's Explanation:

In 1905, Albert Einstein proposed a revolutionary idea: light consists of discrete packets of energy called "quanta" or photons. Each photon has energy proportional to its frequency. When a photon strikes an electron, it transfers its energy directly. If this energy exceeds the work function (the minimum energy needed to eject an electron), the electron is emitted instantly with a kinetic energy proportional to the photon's frequency. Einstein's explanation not only correctly described the photoelectric effect but also provided strong evidence for the quantum nature of light. For this work, Einstein was awarded the Nobel Prize in Physics in 1921.

Photoelectric Effect Simulation App



The above image is a screenshot of an application that can be used to investigate the photoelectric effect. The application is at the following web site.
www.hubbergetter.github.io/tburr-space

Go to the above web address and select *photoelectric effect* from the side menu.

Note that even though it might not look like it, the simulation is running. To see this, move the *Wavelength Slider* back and forth. This will cause the light from the lamp to change color.

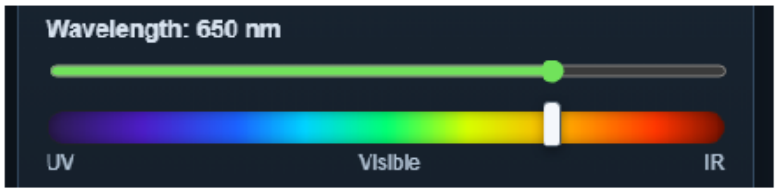
Description of the App

The Emitter and Collector

There are two metal plates that are inside an evacuated tube. One plate is called the *cathode* and the other the *anode*. The plate on the left, on which the lamp is shining, is the cathode. It is negatively charged. It is also called the *emitter* because in some cases, when light shines on it, electrons are emitted. These emitted electrons are attracted to the plate straight across to the right, the anode. The anode is positively charged. The anode is also called the *collector* because it captures those emitted electrons that have enough kinetic energy to reach it.

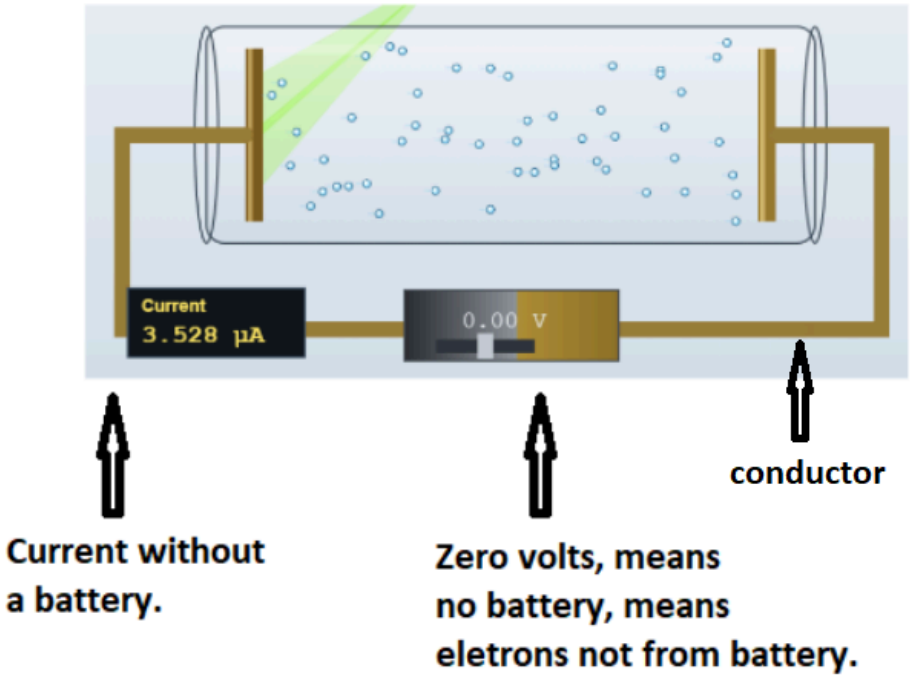
The Lamp

Note that a lamp is used to shine light on the emitter Recall that the color of light is determined by its wavelength. The slider on the top right can be used to adjust the color of the light coming out of the lamp. Move the slider so that the light's wavelength is 650 nm. You should now see a reddish light shining on plate A.



Next, alternately brighten and dim the light beam by using the *Light Intensity* slider that is beneath the *Wavelength slider*.

The Battery and the Ammeter



You will soon see that the above image forms a circuit. When electrons leave the emitter, they travel to the anode, through the conductor and the ammeter, and then back to the cathode.

In parts of this lab the battery will be needed and in other parts it will not. If the *Voltage Slider* is set to zero, this means that the battery will not be needed. In this case, the battery acts just like a conductor.

Emitter Metals

The emitter can be composed of different metals. A metal can be selected from the *Emitter Metal* dropdown menu, which shows just a few emitter metals.

Photoelectric Effect Lab Activity

Heinrich Hertz was the first to observe the photoelectric effect, accidently, in 1887. Hertz was one of the first to investigate this phenomenon. Let's carry out some of the same investigations that he did.

1. Select *Sodium* from the *Emitter Metal Dropdown*.

2. Move the *Wavelength Slider* y to the right and set it at 750 nm.

   a. Change the 750 nm wavelength to frequency. Use c = λf.

   b. What color is the light shining on the emitter?

   c. Are any electrons being emitted from the emitter?

3. Slowly decrease the wavelength of the light. (This is done by moving the *Wavelength Slider* to the left. At some point you will see electrons being emitted from the cathode.
   Stop the slider at this point.

   a. Repeat the above process until you keep getting close to the same answer.
   Then record this wavelength. This wavelength has a special name. It is called the
   threshold wavelength.

   threshold wavelength = _____

b. Usually, physicists use the term *threshold frequency*. Which is denoted by f_0. Find the threshold frequency of sodium. Use c = λf.

c. Find the threshold frequency for all of the metals in the dropdown menu.

i. sodium?

ii. zinc?

iii. copper?

iv. cesium

v. potassium ?

Before Einstein, physicists such as Heinrich Hertz, hypothesized that the photoelectric effect was caused by high intensity light. If light acts only as a wave, as Hertz assumed, then increasing the intensity of a light wave would act the same as increasing the intensity/amplitude of a water wave. A low intensity water wave might not move a log on a beach, but waves with bigger amplitudes, that is higher intensities, could do the job. So he thought that the same is probably true for electrons. That is, that if low intensity infrared light couldn't dislodge an electron from a piece of metal, then higher intensity light should do the job. Also, the effect would not be immediate because, as he thought, the necessary intensity would need time to build up.

Next, you will test Hertz's hypothesis.

4. Move the *Wavelength Slider* to a setting of about 600 nm and the intensity slider to 10%.

a. Are any electrons emitted?

b. Next, increase the intensity to 50%.

i. Is the light brighter?

ii. Are any electrons being emitted?

c. Then increase the intensity to 100%.

ii. Has the intensity increased, i.e. is the light brighter?

iii. Are any electrons being emitted?

iv. What does this tell us about Hertz's hypothesis? Is high intensity light responsible for the photoelectric effect?

5. The above experiment shows that Hertz's hypothesis was incorrect. That the intensity of a light beam does not cause electrons to be emitted. So what is responsible for the photoelectric effect? Earlier you found that the threshold frequency for sodium is about 540 nm.

i. Move the *Intensity Slide*r to a setting of 50%. Then move the *Wavelength Slider* to the left until you see electrons being emitted. What is the *threshold wavelength*?

ii. Keep the *Wavelength Slider* at threshold and move the intensity slider to a setting of around 5%.

- At this low intensity, are any electrons being emitted?

- So what causes the photoelectric effect? ***Explain***.

Albert Einstein was the first person to explain the photoelectric effect, for which he won a Nobel Prize in 1921. His explanation does not rely on classical wave theory. Instead, he proposed that light consists of tiny packets of energy called *photons*, and that electrons are ejected from a metal as a result of the photons colliding into them. The ejected electrons are called ***photoelectrons***, and the current produced is called ***photocurrent***.

Einstein was influenced by a man named Max Planck, who proposed that the energy of a photon is given by the following formula.

$$E = hf$$

- $\boldsymbol{E}$: Energy of the photon (Joules, J)
- $\boldsymbol{h}$: Planck's constant ($6.62607015 \times 10^{-34}$ J · s)
- $\boldsymbol{f}$: Frequency of the electromagnetic radiation (Hertz, Hz or s^{-1})

6. Find the energy of a photon at the different wavelengths.

   a. 350 nm

   b. 500 nm

Finding the Maximum KE of an Ejected Electron

Einstein showed that in order for a photon to eject an electron from a metal, it must have enough energy, $E = hf$, to

1. pull the electron out of the metal; and,
2. provide kinetic energy to the ejected electron.

Electrons are usually trapped in a metal, attracted by the positively charged protons in the metal's atoms. Therefore, part of the photon's energy is used to pull the atom from

the metal. The energy required to accomplish this task is called the metal's work function, denoted by ϕ. Later in the handout, you will see how to find values of ϕ. But first, you will see how to find the maximum kinetic energy of an ejected electron.

The maximum amount of *KE* of an ejected electron can be determined by using the voltage source (the battery) to stop the electrons right before they reach the collector. (Note, not all photoelectrons have enough KE to reach the collector.) To see how this is done for sodium, set the wavelength of the light source to 300 nm and the intensity to 50%. Then use the *Voltage Slider* to slowly make the battery's voltage more and more negative. At about -1.85 volts, some of the electrons should barely reach the right plate and then turn around. When this happens, these electrons are at their maximum *KE*. To see why this happens, consider that when you made the battery negative, the plate on the right took on a negative charge and the one on left took on a positive charge. (This can be seen in the simulation.) This voltage sets up an electric field between the plates that opposes the motion of the electrons that are being ejected from the sodium metal. To stop an electron, work must be done on it and this work/energy must match the electron's ***KE***. That is, we must have

$$KE = qV.$$

So for a wavelength of 300 nm and a stopping voltage of about -1.82, the maximum kinetic energy of an ejected photoelectron from sodium metal is given by

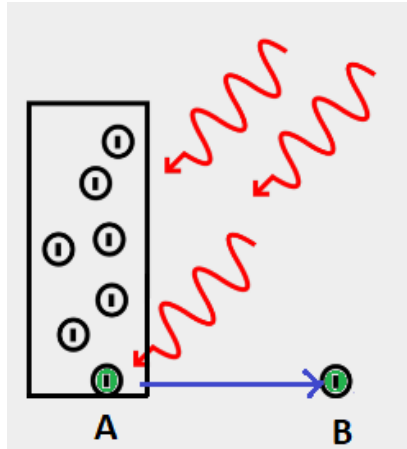
$$\begin{aligned} KE &= qV \\ &= (1.6 \times 10^{-19}\ \mathrm{C})(1.82\ \tfrac{\mathrm{J}}{\mathrm{C}}) \\ &= 2.912 \times 10^{-19}\ \mathrm{J} \\ &= 1.82\ \mathrm{eV} \end{aligned}$$

Earlier it was stated that in order for a photon to eject an electron from a metal, it must have enough energy to

a. pull the electron out of the metal; and,

b. provide kinetic energy to the ejected electron.

We just finished focussing on *item b*. Next, *item a* will be considered. In the following diagram we see each of the following.



a. An electron is bound to the surface of an emitter metal at point A.

b. A photon from the light source strikes the electron.

c. The photon causes the electron to be ejected from the metal and pass through point B.

As we previously saw, the amount of energy that the photon has right before it hits the electron is given by $E = hf$. Some of this energy is used to pull the electron from the metal and the rest becomes the kinetic energy of the ejected electron. As described earlier in this document, the amount of energy used to pull the electron from the metal is called the metal's *work function*, denoted by ϕ.

The mathematic form of the above statement is given by the following equation.

$$hf = KE_{max} + \phi$$

Energy of photon — max kinetic energy — work function

7. Photoelectrons are being emitted from a sheet of metal that is illuminated with 212 nm light.

   a. Find the energy of the incoming photons.

b. Find KE (max)for the photoelectrons.

c. Find the work function, ϕ.

d. Show that the above equation is true. Note that because of experiment error, you might not get the equation to be exactly equal.

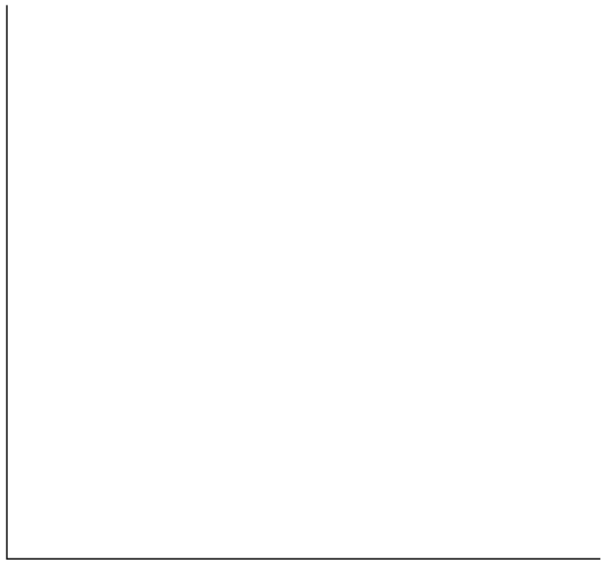
Next, the above equation can be rewritten in the form that shows the maximum kinetic energy of an emitted photon is a function of its frequency. Mathematically, this equation takes the same form as $y = mx + b$. You will take some measurements and come up with a scatter plot consisting of *(f, KE)* ordered pairs. Then use ***linear regression*** to find and graph the equation for the line of best fit.

$$KE_{max} = hf - \phi$$

8. Complete the following table.

wave length, λ, (nm)	frequency, (f)	voltage	Maximum KE, (J)
200			
300			
400			
500			

9. Next, use the data in the previous table to sketch maximum KE as a function of frequency. Be sure to label the axes. Plot points only. For now, don't graph a line.



11. Use linear regression to find the equation for the line of best fit.

12. Graph the equation on the above axes.

13. Record the slope with units. What does this value represent?

14. The slope of the line is Planck's constant, mentioned earlier. Perform a percent error calculation
between the true value and your experimental value, for this constant.

15. Find and interpret the *f* intercept.

16. Find and interpret the *KE* intercept.

The Relationship Between Intensity and Current

Using the simulation, shining light of **λ** =100 nm, at 50% intensity on a piece of sodium.

17. Record the amount of current.

_____

18. Increase the intensity of the ultraviolet light to 75% and record the current.

_____

19. Increase the intensity of the ultraviolet light to 100% and record the current.

_____

20. Write a brief statement describing the relationship between the intensity of the 100 nm light incident on the sodium and the resulting current.

21 Next, repeat the above activity for wave lengths of 200 nm, 300 nm, and 400 nm. Be sure to complete the following table.(This is exercise 15, so do the work on that page.)

wave length, λ, (nm)	Intensity, %			Current, I, (A)		
100	50	75	100			
200	50	75	100			
300	50	75	100			
400	50	75	100			